SKYTECH

Software Testing Presentation

Information Systems

Testing strategy • defects • tools • improvements

Presentation Line-Up

1. Software Description — Noluthando
2. Functions of the Software — Fanele
3. Types of Tests Conducted — Andiswa
4. Software Tools & Techniques Used — Mondli
5. Software Bugs / Defect Identification — Sisanda
6. How the Software Can Be Improved — Vuyani
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1. Software Description

Presenter: Noluthando

SkyTech is a growing South African retailer of new and pre-owned electronics, gadgets and appliances.

The business also provides buy/sell services and loans, supporting affordable access to goods and quickcash solutions.

The software digitizes store operations into an integrated e-commerce platform.

Customers can browse, purchase and track orders online while accessing additional services such as repairs, collections and loans.

2. Functions of the Software

Presenter: Fanele

Product purchasing and payments: browse products, add to cart, pay and receive confirmation.

Order fulfilment: process orders, assign delivery/collection, track progress and complete fulfilment.

Repair services: submit repair requests, track service status, receive updates and schedule pickup.

HR & recruitment: post vacancies, accept applications, test and rank candidates, and schedule interviews.

Staff operations: facial-recognition check-in/check-out and shift-change requests.

Loan services: customers apply for loans and submit assets as collateral.

3. Types of Tests Conducted

TESTING STRATEGY

Presenter: Andiswa

Functional Testing — verified that required functions work correctly, including login, role-based access and the six core business processes.

System Integration Testing — checked that connected sub-systems work together, especially order processing and payment. White-box Testing — developers examined internal code and business rules for discounts, candidate scoring and interviews.

3. Types of Tests Conducted

TESTING STRATEGY

Presenter: Andiswa

Non-Functional Testing — evaluated security, performance and compatibility across phones, tablets and laptops.

Security example: facial recognition should prevent one staff member from using another person's identity to start or end a shift.

Performance example: the system should remain responsive when many applicants submit job applications.

Black-box Testing — testers supplied inputs and compared outputs with requirements without focusing on the internal code.

Example: a candidate who fails the assessment must not be allowed to proceed to an interview.

4. Software Tools & Techniques Used

STATIC TESTING

Presenter: Mondli

Review Analysis: informal reviews, walkthroughs and formal inspections were used to find problems in requirements, designs and code before execution.

Checkstyle: checked Java/mobile code against coding standards to improve readability and maintainability.

SonarQube: analysed code for bugs, code smells, vulnerabilities, duplication and complexity; applied to loan calculations and candidate scoring.

Equivalence Partitioning: grouped loan-income inputs into representative ranges to reduce unnecessary test cases.



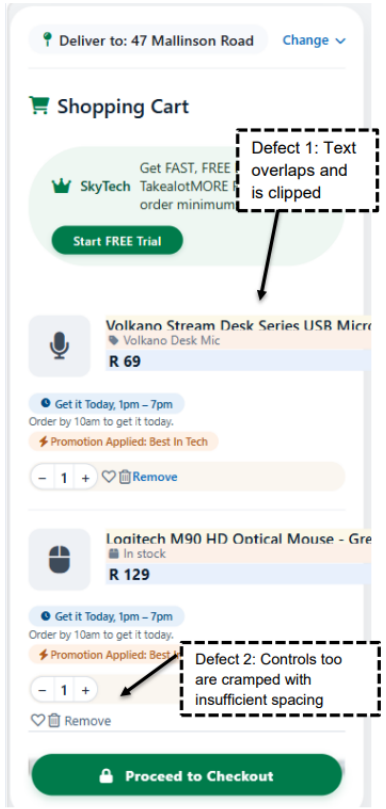
Decision Table Testing: combined candidate conditions such as internet stability, test completion and pass mark to determine the correct interview outcome.

	TC 1	TC 2	TC3	TC4	TC5	TC6	TC7	TC8
Conditions								
Has stable internet	True	False	True	False	True	True	False	False
Completed the test	True	True	False	False	True	False	False	True
Got required pass mark	True	True	False	False	False	True	True	False
Actions								
Invited for online interview	True	False	False	False	False	True	False	False
Invited for physical interview	False	True	False	False	False	False	True	False
Not invited for interview	False	False	True	True	True	False	False	True

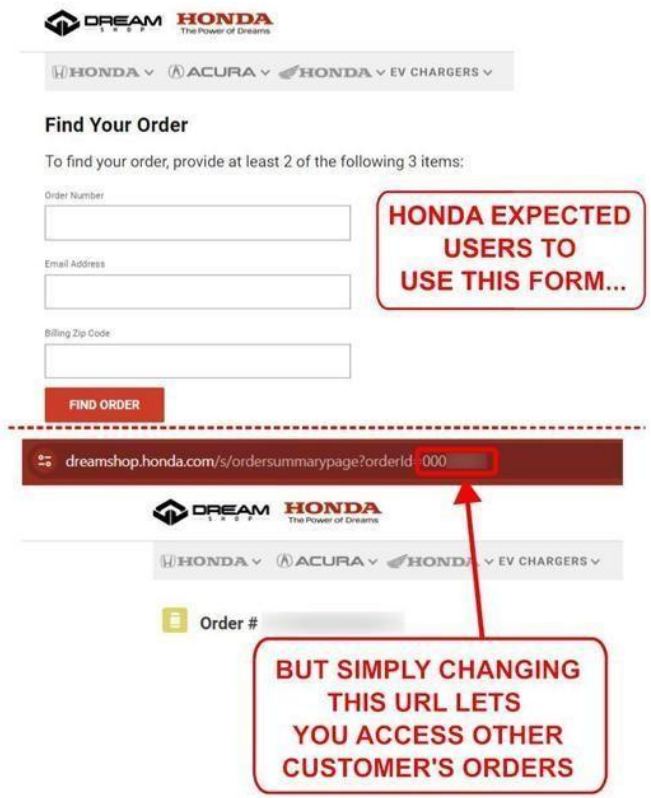
5. Software Bugs / Defect Identification

Presenter: Sisanda

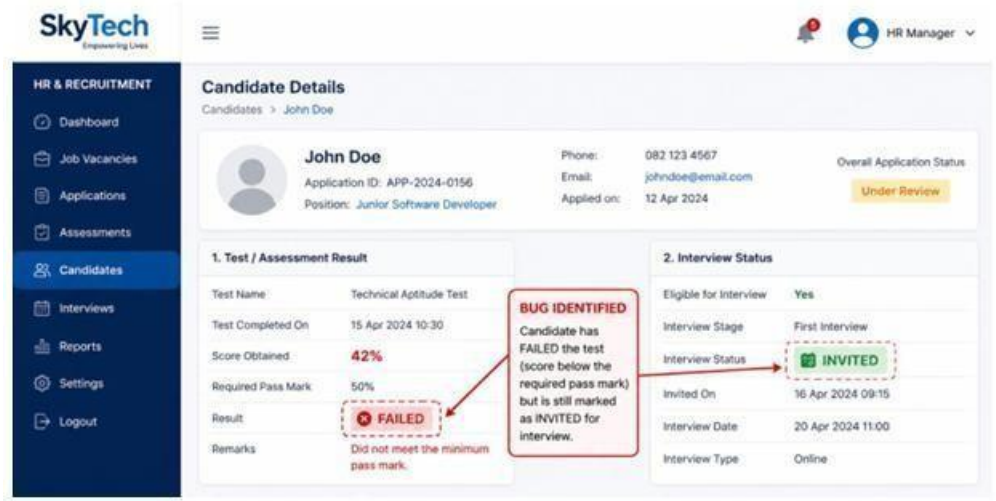
BUG 001 — GUI: checkout elements overlap on mobile screens. Severity: High.


Deliver to: 47 Mallinson Road
Change
Shopping Cart
Defect 1: Text overlaps and is clipped
SkyTech
Get FAST, FREE
TakealotMORE
order minimum
Start FREE Trial
Volkano Desk Mic
R 69
Get it Today, 1pm – 7pm
Order by 10am to get it today.
Promotion Applied: Best In Tech
1
Remove
In stock
R 129
Get it Today, 1pm – 7pm
Order by 10am to get it today.
Defect 2: Controls too are cramped with insufficient spacing
Remove
Proceed to Checkout

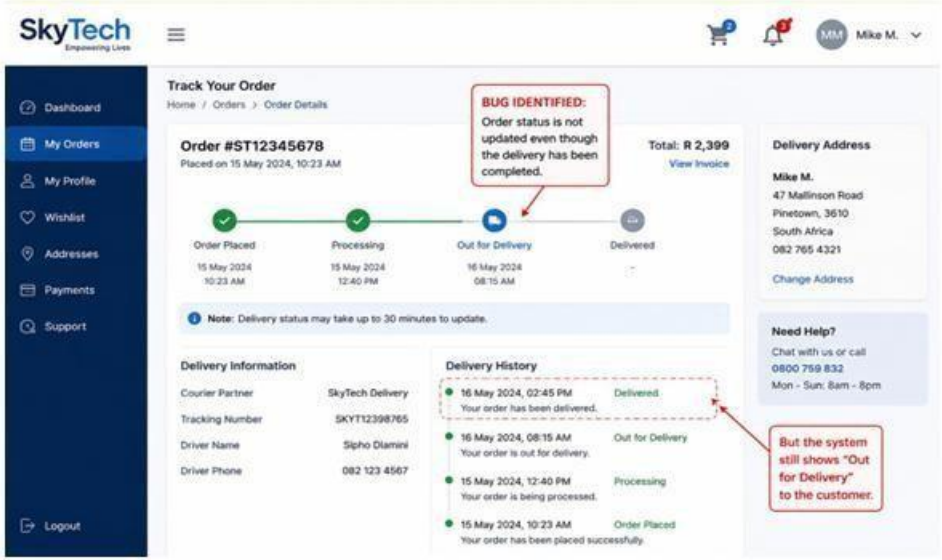
BUG 002 — Security/Authorization: a customer can access another customer's order information. Severity: Critical.


DREAM SHOP HONDA The Power of Dreams
HONDA ACURA HONDA EV CHARGERS
Find Your Order
To find your order, provide at least 2 of the following 3 items:
Order Number
Email Address
Billing Zip Code
FIND ORDER
HONDA EXPECTED USERS TO USE THIS FORM...
dreamshop.honda.com/s/ordersummarypage?orderId=000
DREAM SHOP HONDA The Power of Dreams
HONDA ACURA HONDA EV CHARGERS
Order #
BUT SIMPLY CHANGING THIS URL LETS YOU ACCESS OTHER CUSTOMER'S ORDERS

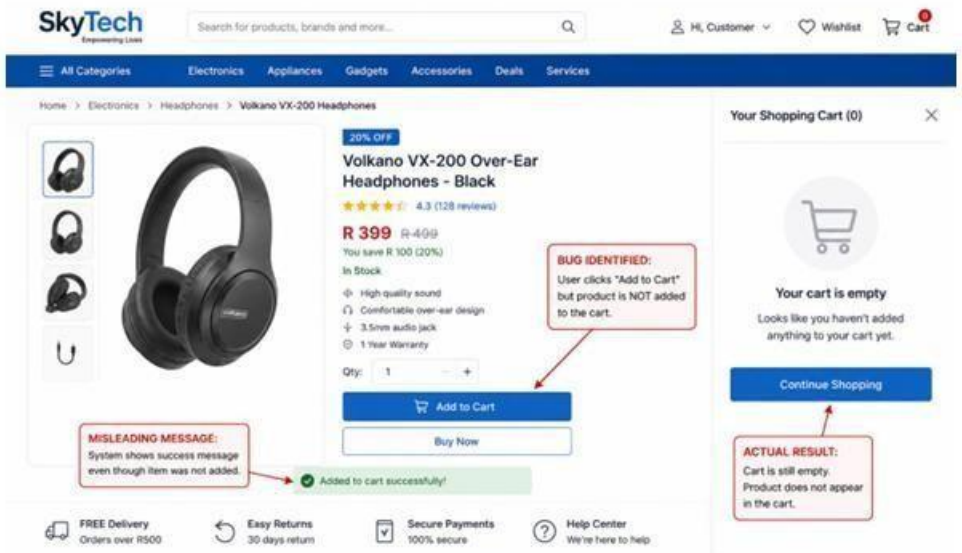
BUG 003 — Recruitment: a failed candidate can still be invited for an interview. Severity: Critical.



BUG 004 — Delivery: customer tracking displays an outdated delivery status. Severity: High.



BUG 005 — Add to Cart: the button appears successful, but the product is not added. Severity: Critical.


SkyTech
Search for products, brands and more...
Hi, Customer
Wishlist
Cart
All Categories
Electronics
Appliances
Gadgets
Accessories
Deals
Services
Home > Electronics > Headphones > Volkano VX-200 Headphones
20% OFF
Volkano VX-200 Over-Ear Headphones - Black
4.3 (128 reviews)
R 399 R 499
You save R 100 (20%)
In Stock
High quality sound
Comfortable over-ear design
3.5mm audio jack
1 Year Warranty
Qty: 1
Add to Cart
Buy Now
BUG IDENTIFIED:
User clicks "Add to Cart" but product is NOT added to the cart.
MISLEADING MESSAGE:
System shows success message even though item was not added.
Added to cart successfully!
Your Shopping Cart (0)
Your cart is empty
Looks like you haven't added anything to your cart yet.
Continue Shopping
ACTUAL RESULT:
Cart is still empty. Product does not appear in the cart.
FREE Delivery
Orders over R500
Easy Returns
30 days return
Secure Payments
100% secure
Help Center
We're here to help

5. Why These Defects Matter

DEFECT IMPACT

Presenter: Sisanda

GUI defects can block mobile customers from completing checkout and reduce usability.

Broken access control exposes customer information and creates a serious privacy/security risk.

Recruitment logic defects undermine fairness by allowing candidates who failed the test to proceed.

Incorrect delivery status reduces trust and can cause confusion or disputes.

Cart failures directly prevent purchases and can result in lost sales.

6. How the Software Can Be Improved

RECOMMENDATIONS

Presenter: Vuyani

Improve responsive UI design so checkout fields, buttons and icons display correctly on different screen sizes.

Strengthen role-based authentication and authorization so users can access only information they are permitted to view.

Enforce backend validation for recruitment pass-mark rules and prevent failed candidates from moving to interview stages.

Synchronize delivery status across the UI and database, with audit logs for delivery date/time changes.

Use automated regression testing with tools such as Selenium WebDriver and Postman to confirm fixes and prevent defects from returning.

7. Conclusion

CLOSING

Presenter: Sihle

SkyTech has a strong foundation as an integrated digital platform for retail and supporting services.

Testing exposed five significant defects affecting usability, security, recruitment fairness, delivery accuracy and purchasing.

Functional, non-functional, white-box and black-box testing helped identify weaknesses from different perspectives.

Applying stronger authorization, responsive design, data synchronization and automated regression testing will improve reliability, security and scalability.

QUESTIONS?